

DONALD B. HAVERY

Applied AI Systems Engineer · AI systems engineered for production · Agentic Orchestration · RAG & Context Engineering · Eval Gates · Security Controls · MCP · Local-First

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SUMMARY

Applied AI systems engineer. I build the engineering around the models that gets them into production: agent orchestration, RAG, eval gates, and security controls. The fastest proof is public and measurable. Three MIT, CI-green builds on GitHub: an **LLM reliability gateway** with cross-provider failover (115 tests), an **evaluation release gate** that exits non-zero on regression (54 tests), and an **agent orchestration reliability harness** with typed tools, bounded retries, and an unsafe-action gate (45 tests), plus a published paper on reference-relative believability (Zenodo DOI). Broader work: **Companion** (local-first voice assistant; 7-provider model router; 12-persona system), **Albums** (RAG over ~3,600 indexed items), and **Wire** (an unattended pipeline that researched and published a daily brief, 35 posts in production). 38 repos, 3,300+ commits, 5,000+ tests authored; heavy Claude Code user with 25+ skills of my own. Before software I founded and ran my own businesses end-to-end and certified new operators to a federal standard inside safety-critical, DOT-regulated work, so I build software already knowing what it has to do for the business that depends on it.

REPRESENTATIVE PROJECTS

An LLM reliability gateway with cross-provider failover (Citadel)

Python · FastAPI · HNSW (pure NumPy) · 115 tests · MIT · public · CI green · github.com/dbhavery/citadel

- An OpenAI-compatible control layer for LLM traffic: provider priorities with real **cross-provider failover**, circuit breakers, semantic cache, and a from-scratch **HNSW vector engine** in pure NumPy (built from Malkov & Yashunin 2018; no FAISS / pgvector). Public code, MIT, green CI, cloneable tests

An evaluation release gate for LLM & RAG behavior (eval-gate)

Python · 54 tests · MIT · public · CI green · github.com/dbhavery/eval-gate

- Prompt assertions, retrieval precision / recall / MRR / nDCG, citation grounding, and baseline comparison — runs offline and deterministic, then **exits non-zero when a case regresses**, so a bad model change fails the build instead of reaching production

An agent orchestration reliability harness (agent-harness)

Python · 45 tests · MIT · public · CI green · github.com/dbhavery/agent-harness

- A deterministic agent loop built around production failure modes: typed tool schemas, structured JSONL traces, bounded retries and fallback, an **unsafe-action gate**, and explicit failure classification (timeout, malformed output, missing context, unsafe request)

A local-first voice assistant with a 7-provider model router (Companion)

Python · Rust + Tauri · PySide6 · Android · 695 commits · 345 tests

- **7-provider model router** (Claude, Gemini, Qwen, local Ollama, OpenAI, Groq, OpenRouter) with 3-tier complexity routing — regex, then keyword, then an LLM classifier — choosing the right model per request
- **Context engineering**: ChromaDB hybrid retrieval (BM25 + dense embeddings), tiered memory, MCP integration, async tool execution; **12-persona system** with consistent behavior and failover across providers; sub-500 ms latency budget

Semantic file library with RAG chat, and an unattended research-to-publish pipeline (Albums & Wire)

FastAPI · React · ChromaDB · Python cron

- **Albums** — self-hosted Google Photos / Drive replacement: semantic search and RAG chat over ~3,600 indexed media items, the same retrieval patterns Companion uses on a larger corpus
- **Wire** — a self-reviewing pipeline that pulled ~18 sources, drafted, fact-checked, generated a hero image, and auto-published a daily brief with no human in the loop; 35 posts in production before I parked it

Additional public builds & reusable tooling

MIT · public · github.com/dbhavery

- **control-evidence-review** (policy/control → evidence verdicts with redaction and an append-only audit log, 40 tests), **text-to-CAD** (local plain-English → OpenSCAD loop with a vision-model repair pass, 23 tests), **Graticule** (live-data situational-awareness globe). 25+ reusable Claude Code skills and agent specs, each with explicit refusal boundaries and verification gates

TECHNICAL SKILLS

Agentic & Tool Use	Multi-agent orchestration , sub-agent coordination, intent routing, autonomous loops, tool use / function calling at scale, MCP (production integrations, custom servers), behavioral specs, fallback chains
RAG & Retrieval	Hybrid retrieval (BM25 + dense), ChromaDB, from-scratch HNSW, sentence-transformer embeddings, tiered memory, context engineering
Evals & Reliability	Deterministic offline assertions , adversarial & canary probes, meta-evaluation, regression detection, CI exit-code gating; circuit breakers, bulkhead isolation, retry budgets, observability for AI
Claude Code	Heavy Claude Code user — 3,300+ commits authored through it, 25+ skills and custom agents of my own, plugin development
LLM Stack	Claude (Opus/Sonnet/Haiku), Gemini, Qwen, Ollama local serving; cross-provider parity, latency budgets; LoRA / QLoRA fine-tuning (image-domain LoRA shipped), HuggingFace, PyTorch
Backend & Product	Python (asyncio, FastAPI), Rust (Tauri, tokio), Node.js, PostgreSQL / SQLite / Redis, Docker, GitHub Actions; Next.js, React 19, TypeScript, PySide6 desktop
Operations & Ventures	Founder / owner-operator : business operations, DTC e-commerce, product formulation & small-batch manufacturing, brand & marketing, supply chain & fulfillment; investigative case management; HAZMAT-certified regulated transport (DOT/FMCSA), risk & safety management

EXPERIENCE

Applied AI Systems Engineer — Independent

Current

- Design and ship production agentic systems end-to-end: orchestration, tools, retrieval, evaluation, security controls, fine-tuning. 38 repos, 3,300+ commits, 5,000+ automated tests authored
- Published three MIT, CI-green proof builds — an LLM reliability gateway, an eval release gate that fails the build on regression, and an agent reliability harness — each with cloneable tests and inspectable reports
- Operate as a Claude Code pair-programmer daily; keep prompts, evals, and harnesses in version control and under test, the same as any other production code
- Public portfolio at dbhavery.dev leads with the three flagship builds, real screenshots, and a published research paper; every flagship card opens a public repo you can clone and test

Founder & Owner-Operator — Independent Ventures

- **Original Lumberjack** — premium men's grooming, DTC e-commerce. Ran the whole business solo: formulated products from raw ingredients, manufactured in small batches, designed the brand, packaging, and labeling, and handled marketing, fulfillment, shipping, and customer service
- **Precision Investigations** — independent case work: investigative research, surveillance, case management, evidence handling, client reporting, and strict confidentiality
- **Sta-Full** (fleet-fueling logistics) and **angrybirds** (a remote, contract AI-engineering collective, currently building) — scoped, branded, and stood up independently
- Every venture built and run end-to-end by one person, alongside full-time work

Operations & Safety-Critical Logistics — Certified Trainer

Federally regulated, high-consequence operations under DOT/FMCSA oversight: HAZMAT-certified, zero-error standards, real liability. Certified to train and sign off new operators to a federal standard, with documented process and clear accountability when something went wrong. That is where my verification habit comes from, and it carries directly into how I build and run production AI.

PUBLICATIONS

“Reference-Relative Believability” — published research paper introducing a reference-relative method for evaluating the perceived believability of AI-generated media. DOI 10.5281/zenodo.21138188 (Zenodo).

EDUCATION

Capella University — B.S. Computer Science (in progress).

CERTIFICATIONS

Anthropic Academy — 15 courses complete (Building with Claude, Claude Code, Claude Code Skills, MCP & MCP Advanced, Cowork, Bedrock, Vertex AI, AI Fluency suite, AI Capabilities & Limitations)

DeepLearning.AI — Generative AI with Large Language Models | **Google** — AI Fundamentals (Coursera-verifiable)

Sophia Learning — 30 ACE college-credit courses, verified (public Credly profile) — CS foundations (Python, Java, web dev, SQL/databases, networking, statistics) through business and analytics